

Positive LDO (LT3042): Working Document

Digital Audio Player • Rev 8

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1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LT3042EDD#TRPBF
Package	10-lead DFN, 3 × 3 mm, exposed pad (pin 11) = GND
Function	Ultralow-noise positive LDO, +3V9 → +3.3 V analog rail

2. Design Requirements

Requirement	Value
Input	+3V9 from LTC3440, 3.827 to 3.981 V; 7.7 mV pp ripple; ±90 mV load step (minimum 3.74 V)
Output	3.32 V
Load	I/V, summing, OPA1622 (V+ side), DAC analog supplies
Maximum audio output	0.6 V RMS (EW300: 119 dB/V worst nozzle, 28 Ω ±15%)
Output current	≈ 35 mA quiescent (OPA1612 21.6 mA, OPA1622 5.2 mA, DAC analog 8 mA); ≈ 106 mA peak (200 mA rated)
Current limit	125 mA programmed (R_{ILIM})
UVLO	3.5 V rising
Power good	≈ 91% of V_{OUT} , to MCU GPIO
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1, 2	IN	+3V9, C_{IN}
3	EN/UV	R_{EN2} from IN, R_{EN1} to GND
4	PG	R_{PU} to +3V3, MCU GPIO
5	ILIM	R_{ILIM} to GND (Kelvin to pin 8)
6	PGFB	R_{PG2} / R_{PG1} midpoint
7	SET	$R_{SET} \parallel C_{SET}$ to GND; C_{SET} anode to SET
8	GND	GND
9	OUTS	OUT at the load
10	OUT	+3V3A, C_{OUT} , R_{PG2}
11	Exposed pad	GND

4. External Components

Ref	Value / Rating	MPN	AL-PN
R_{SET}	33.2 k Ω 0.1% 0402	YAGEO RT0402BRD0733K2L	#1-085
C_{SET}	2.2 μ F $\pm$ 20% tantalum, 16 V, 0603, polarized (no DC-bias derating, not piezoelectric)	Vishay 298D225X0016M2T	#1-098
C_{IN}	10 μ F 10 V X7R 0805 (6.76 μ F effective at 3.9 V)	Murata GCM21BR71A106KE22K	#1-081
C_{OUT}	10 μ F 10 V X7R 0805 (7.56 μ F effective at 3.3 V)	Murata GCM21BR71A106KE22K	#1-081
R_{EN1}	50 k Ω 0.1% 0603	YAGEO RT0603BRD0750KL	#1-075
R_{EN2}	90.9 k Ω 0.1% 0402	YAGEO RT0402BRD0790K9L	#1-084
R_{PG1}	100 k Ω 0.1% 0603	YAGEO RT0603BRD07100KL	#1-066
R_{PG2}	909 k Ω 1% 0402	YAGEO RC0402FR-07909KL	#1-083
R_{ILIM}	1.0 k Ω 0.1% 0402	YAGEO RT0402BRD071KL	
R_{PU}	10 k Ω 0603	Generic	

5. Equations and Results

Output voltage

$$V_{OUT} = I_{SET} R_{SET} = 100 \mu A \times 33.2 k\Omega = 3.32 V$$

Source: [LT3042 Datasheet \(Analog Devices\)](#), SET pin.

Undervoltage lockout (I_{EN} term neglected)

$$V_{IN(UVLO)} = 1.24 \left(1 + \frac{R_{EN2}}{R_{EN1}} \right) = 1.24 \left(1 + \frac{90.9}{50} \right) = 3.494 V$$

Source: [LT3042 Datasheet \(Analog Devices\)](#), EN/UV pin.

Power-good threshold (I_{PGFB} term neglected)

$$V_{OUT(PG)} = 0.3 \left(1 + \frac{R_{PG2}}{R_{PG1}} \right) = 0.3 \left(1 + \frac{909}{100} \right) = 3.027 \text{ V (91.2\%)}$$

Source: [LT3042 Datasheet \(Analog Devices\)](#), PGFB pin.

Current limit (ILIM threshold 300 mV; 125 mA·kΩ scale factor)

$$I_{LIM} = \frac{125 \text{ mA} \cdot \text{k}\Omega}{R_{LIM}} = \frac{125 \text{ mA} \cdot \text{k}\Omega}{1.0 \text{ k}\Omega} = 125 \text{ mA} \quad (1.18 \times \text{the } 106 \text{ mA peak})$$

Source: [LT3042 Datasheet \(Analog Devices\)](#), ILIM pin.

SET noise-filter corner (1 μF effective)

$$f_c = \frac{1}{2\pi R_{SET} C_{SET}} = \frac{1}{2\pi \cdot 33.2 \text{ k}\Omega \cdot 1 \mu\text{F}} = 4.8 \text{ Hz}$$

Soft-start (fast start-up to ≈ 2.7 V, then the $R_{SET} C_{SET}$ tail; 1 μF effective)

$$\tau = R_{SET} C_{SET} = 33.2 \text{ ms}, \quad t_{99} = \tau \ln \frac{0.62}{0.033} = 2.93 \tau \approx 97 \text{ ms}/\mu\text{F}$$

Source: [LT3042 Datasheet \(Analog Devices\)](#), fast start-up.

Headroom and maximum SPL

$$V_{IN(MIN)} - V_{OUT} = 3.74 - 3.32 = 0.42 \text{ V}$$
$$\text{SPL}_{MAX} = 119 + 20 \log_{10}(0.6) = 114.6 \text{ dB (116.6 dB steel nozzle)}$$

C_{SET} sweep (simulated; LTC3440 replaced by an ideal 3.9 V source; 29 Ω load):

C_{SET} (effective)	t_{99}	Noise, 20 Hz to 20 kHz
0.47 μF	46 ms	1.16 μV RMS
1 μF	98 ms	0.63 μV RMS
2.2 μF	214 ms	0.40 μV RMS
4.7 μF	455 ms	0.34 μV RMS

Load current (0.6 V RMS = 0.85 V peak design maximum, both channels in phase):

Case	Per channel pk	Signal total	+ Quiescent	Regulator pk
Full scale, 23.8 Ω (28 Ω –15%)	35.7 mA	71.3 mA	35 mA	≈ 106 mA
Full scale, 28 Ω nominal	30.3 mA	60.6 mA	35 mA	≈ 96 mA
110 dB SPL peak (0.35 V RMS), 23.8 Ω	21.1 mA	42.2 mA	35 mA	≈ 77 mA

6. Important Values

Item	Value
SET pin current	100 μ A
EN/UV rising threshold	1.24 V
PGFB threshold	300 mV
Dropout	350 mV typ at 200 mA
Rated output current	200 mA
Current limit	125 mA programmed; ILIM also monitors output current (1:400, 0 to 300 mV)
Peak load	$\approx$ 106 mA at 0.6 V RMS maximum (53% of rating)
Minimum headroom	0.42 V
UVLO / PG divider current	$\approx$ 28 μ A from +3V9 / $\approx$ 3.3 μ A from OUT
Output settled (99%)	$\approx$ 214 ms after enable (2.2 μ F, simulated)
Output noise	0.40 μ V RMS, 20 Hz to 20 kHz (2.2 μ F, simulated)
C_{SET} leakage	0.5 μ A max at 16 V rating ($\leq$ 0.5% output error); far lower at 3.3 V
Firmware delay	Wait $\approx$ 214 ms after enable before enabling the OPA1622
PG in shutdown	High impedance: reads good while EN/UV is low (simulated)

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-21	Initial issue: R_{SET} , UVLO, PG, soft-start, load current.
1	2026-09-22	C_{SET} 2.2 μ F nominal; start-up and noise sweeps.
2	2026-09-22	C_{IN} / C_{OUT} 10 μ F GCM21BR71A106KE22K.
3	2026-09-22	Library parts assigned; R_{PG2} to 1%.
4	2026-09-22	Maximum output 0.6 V RMS; peak load 98 mA.
5	2026-09-22	Current limit 125 mA: R_{ILIM} 1.0 k Ω .
6	2026-09-23	C_{SET} 2.2 μ F tantalum 298D225X0016M2T, anode to SET.
7	2026-09-25	Reformatted; linked contents and sources; open problems closed.
8	2026-09-27	Load updated from Power Budget Rev 6: quiescent $\approx$ 35 mA, peak $\approx$ 106 mA (current limit 1.18 $\times$ the peak).

9. Sources

1. [LT3042 Datasheet \(Analog Devices\)](#)